

Qualifying Function of One Complex Variable I

John B. Conway (Second Edition)

Summer 2026-2027

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Chapter 1

August 2018

1.1 Part I

1. *Find a conformal map f from the vertical strip*

$$\Omega = \{z \in \mathbb{C} : 3 < \operatorname{Re}(z) < 11\}$$

onto the open unit disk $\mathbb{D}$.

Solution. The width of the strip is $11 - 3 = 8$. First map the strip to a horizontal strip by setting

$$w = \frac{\pi}{8}(z - 3).$$

Since $3 < \operatorname{Re}(z) < 11$, we have

$$0 < \operatorname{Re}(w) < \pi.$$

Now define

$$\omega = e^{iw}.$$

Because

$$\arg(\omega) = \operatorname{Re}(w),$$

the map $\omega = e^{iw}$ sends the strip onto the upper half-plane

$$\{\omega \in \mathbb{C} : \operatorname{Im}(\omega) > 0\}.$$

Finally, the map

$$\phi(\omega) = \frac{\omega - i}{\omega + i}$$

maps the upper half-plane conformally onto the unit disk $\mathbb{D}$. Therefore a conformal map from Ω onto $\mathbb{D}$ is

$$f(z) = \frac{e^{i\pi(z-3)/8} - i}{e^{i\pi(z-3)/8} + i}.$$

■

2. Suppose that F and G are holomorphic functions on the right half plane

$$H = \{z \in \mathbb{C} : \operatorname{Re}(z) > 0\}.$$

Suppose also that F and G have the same zeros with the same multiplicities at each zero. Prove that there is a holomorphic function h on H such that

$$F(z) = e^{h(z)}G(z)$$

for all $z \in H$.

Solution. If $F \equiv G \equiv 0$, then we may take $h \equiv 0$. Thus assume that F and G are not identically zero.

Define

$$q(z) = \frac{F(z)}{G(z)}$$

at points where $G(z) \neq 0$. We claim that q extends to a holomorphic function on all of H .

Let $a \in H$ be a zero of G . Since F and G have the same zeros with the same multiplicities, there is some $m \geq 1$ such that locally

$$F(z) = (z - a)^m \phi(z), \quad G(z) = (z - a)^m \psi(z),$$

where ϕ and ψ are holomorphic near a , with

$$\phi(a) \neq 0, \quad \psi(a) \neq 0.$$

Hence near a ,

$$q(z) = \frac{F(z)}{G(z)} = \frac{\phi(z)}{\psi(z)}.$$

So the singularity of q at a is removable. Therefore q extends holomorphically to all of H .

Moreover, the extension of q has no zeros. Indeed, away from the zeros of G , this is clear since F and G have the same zeros. At a common zero a , we have

$$q(a) = \frac{\phi(a)}{\psi(a)} \neq 0.$$

Now $H = \{z \in \mathbb{C} : \operatorname{Re}(z) > 0\}$ is convex, hence simply connected. Since q is holomorphic and never zero on H , by Corollary 6.17, Chapter IV, there exists a holomorphic function h on H such that

$$q(z) = e^{h(z)}.$$

Therefore, for all $z \in H$,

$$F(z) = q(z)G(z) = e^{h(z)}G(z).$$

Thus there is a holomorphic function h on H satisfying

$$F(z) = e^{h(z)}G(z)$$

for all $z \in H$. ■

3. Suppose f is a holomorphic function on the open unit disc

$$D = \{z : |z| < 1\}.$$

Prove that there is a sequence $(z_n)_{n=1}^{\infty}$ in D such that $|z_n| \rightarrow 1$ and the set

$$\{f(z_n) : n \geq 1\}$$

is bounded.

Solution (I). If $f \equiv 0$, then take $z_n = 1 - \frac{1}{n}$. Then $|z_n| \rightarrow 1$ and $\{f(z_n) : n \geq 1\} = \{0\}$, which is bounded.

Now assume $f \not\equiv 0$. Suppose, toward a contradiction, that no such sequence

exists. Then for every $M > 0$, there is $0 < \rho < 1$ such that

$$|f(z)| > M$$

whenever $\rho < |z| < 1$. Indeed, otherwise we could find $z_n \in D$ with

$$|z_n| > 1 - \frac{1}{n} \quad \text{and} \quad |f(z_n)| \leq M,$$

which would give $|z_n| \rightarrow 1$ and $\{f(z_n)\}$ bounded.

Thus

$$|f(z)| \rightarrow \infty \quad \text{as } |z| \rightarrow 1.$$

In particular, for some $0 < R < 1$,

$$|f(z)| > 1$$

whenever $R < |z| < 1$. Hence all zeros of f lie in the compact set $\{|z| \leq R\}$. Since $f \not\equiv 0$, it has only finitely many zeros there. Let those zeros be $\alpha_1, \dots, \alpha_N$, with multiplicities $m_1, \dots, m_N$, and define

$$P(z) = \prod_{j=1}^N (z - \alpha_j)^{m_j}.$$

If f has no zeros, take $P \equiv 1$.

Then

$$F(z) = \frac{f(z)}{P(z)}$$

extends to a holomorphic function on D , and F has no zeros in D . Therefore

$$g(z) = \frac{1}{F(z)}$$

is holomorphic on D .

Since P is bounded on D , and $|f(z)| \rightarrow \infty$ as $|z| \rightarrow 1$, we get

$$|F(z)| \rightarrow \infty \quad \text{as } |z| \rightarrow 1.$$

Thus

$$g(z) \rightarrow 0 \quad \text{as } |z| \rightarrow 1.$$

Fix $z_0 \in D$. Given $\varepsilon > 0$, choose $\rho < 1$ such that

$$|g(z)| < \varepsilon$$

whenever $\rho < |z| < 1$. Choose r such that

$$\max\{|z_0|, \rho\} < r < 1.$$

By the maximum modulus principle applied on $\{|z| \leq r\}$,

$$|g(z_0)| \leq \max_{|z|=r} |g(z)| < \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, $g(z_0) = 0$. Hence $g \equiv 0$ on D , which is impossible because $g = 1/F$.

Therefore our supposition was false. Hence there exists a sequence $(z_n) \subset D$ such that

$$|z_n| \rightarrow 1$$

and

$$\{f(z_n) : n \geq 1\}$$

is bounded. ■

Solution (II). Suppose, toward a contradiction, that no such sequence exists. Then for every $M > 0$, there exists $0 < \rho < 1$ such that

$$|f(z)| > M$$

whenever $\rho < |z| < 1$. Otherwise, for some $M > 0$, we could choose $z_n \in D$ with

$$|z_n| > 1 - \frac{1}{n} \quad \text{and} \quad |f(z_n)| \leq M,$$

which would give the desired sequence.

Thus

$$|f(z)| \rightarrow \infty \quad \text{as } |z| \rightarrow 1.$$

In particular, there exists $0 < R < 1$ such that

$$|f(z)| > 1$$

whenever $R < |z| < 1$. Hence f has no zeros in the annulus

$$A = \{z : R < |z| < 1\}.$$

Therefore

$$g(z) = \frac{1}{f(z)}$$

is holomorphic on A . Also,

$$g(z) \rightarrow 0 \quad \text{as } |z| \rightarrow 1.$$

Now expand g in its Laurent series on the annulus A :

$$g(z) = \sum_{n=-\infty}^{\infty} c_n z^n.$$

For any r with $R < r < 1$,

$$c_n = \frac{1}{2\pi i} \int_{|\omega|=r} \frac{g(\omega)}{\omega^{n+1}} d\omega.$$

Hence

$$|c_n| \leq \max_{|\omega|=r} |g(\omega)| r^{-n}.$$

Letting $r \rightarrow 1^-$, we have

$$\max_{|\omega|=r} |g(\omega)| \rightarrow 0,$$

and $r^{-n} \rightarrow 1$. Therefore

$$c_n = 0$$

for every $n \in \mathbb{Z}$. Hence

$$g \equiv 0$$

on A , which is impossible since $g = 1/f$.

Therefore our supposition was false. Thus there exists a sequence $(z_n) \subset D$ such that

$$|z_n| \rightarrow 1$$

and

$$\{f(z_n) : n \geq 1\}$$

is bounded. ■

4. Determine, with proof, all entire functions f such that for all $\lambda \in \mathbb{C}$ with $|\lambda| = 1$ and all $z \in \mathbb{C}$ the equation

$$f(\lambda z) = \lambda f(z)$$

holds.

Solution (I). We claim that the only such functions are

$$f(z) = cz$$

for some constant $c \in \mathbb{C}$.

Let f be entire. Then f has a power series expansion on all of $\mathbb{C}$:

$$f(z) = \sum_{n=0}^{\infty} a_n z^n.$$

For $|\lambda| = 1$, we have

$$f(\lambda z) = \sum_{n=0}^{\infty} a_n (\lambda z)^n = \sum_{n=0}^{\infty} a_n \lambda^n z^n.$$

By assumption,

$$f(\lambda z) = \lambda f(z) = \lambda \sum_{n=0}^{\infty} a_n z^n = \sum_{n=0}^{\infty} \lambda a_n z^n.$$

Therefore, for every $|\lambda| = 1$,

$$\sum_{n=0}^{\infty} a_n \lambda^n z^n = \sum_{n=0}^{\infty} \lambda a_n z^n.$$

By uniqueness of power series coefficients,

$$a_n \lambda^n = \lambda a_n$$

for every $n \geq 0$ and every $|\lambda| = 1$. Hence

$$a_n (\lambda^n - \lambda) = 0.$$

If $n = 1$, this condition gives no restriction on a_1 . But if $n \neq 1$, we can choose $|\lambda| = 1$ such that

$$\lambda^n \neq \lambda.$$

Thus $a_n = 0$ for every $n \neq 1$. Therefore

$$f(z) = a_1 z.$$

Conversely, every function of the form $f(z) = cz$ satisfies

$$f(\lambda z) = c\lambda z = \lambda cz = \lambda f(z).$$

Hence the entire functions satisfying the condition are exactly

$$\boxed{f(z) = cz, \quad c \in \mathbb{C}.}$$

■

Solution (II). First put $z = 0$. Then

$$f(0) = \lambda f(0)$$

for every $|\lambda| = 1$. Choosing $\lambda \neq 1$, we get

$$f(0) = 0.$$

Hence

$$g(z) = \frac{f(z)}{z}$$

is holomorphic on $\mathbb{C} \setminus \{0\}$, and the singularity at 0 is removable. Define

$$g(0) = f'(0).$$

Then g is entire.

For $z \neq 0$ and $|\lambda| = 1$,

$$g(\lambda z) = \frac{f(\lambda z)}{\lambda z} = \frac{\lambda f(z)}{\lambda z} = \frac{f(z)}{z} = g(z).$$

Thus, by continuity,

$$g(\lambda z) = g(z)$$

for all $z \in \mathbb{C}$ and all $|\lambda| = 1$.

In particular, if $|\zeta| = 1$, then $\zeta = \lambda \cdot 1$ for some $|\lambda| = 1$, so

$$g(\zeta) = g(1).$$

Thus g is constant on the unit circle.

Write the Taylor expansion of g at 0:

$$g(z) = \sum_{n=0}^{\infty} a_n z^n.$$

By Cauchy's integral formula for Taylor coefficients,

$$a_n = \frac{1}{2\pi i} \int_{|\zeta|=1} \frac{g(\zeta)}{\zeta^{n+1}} d\zeta.$$

For $n \geq 1$, since $g(\zeta) = g(1)$ on $|\zeta| = 1$, we get

$$a_n = \frac{g(1)}{2\pi i} \int_{|\zeta|=1} \frac{1}{\zeta^{n+1}} d\zeta = 0.$$

Therefore all Taylor coefficients of g except possibly a_0 vanish. Hence g is constant on $\mathbb{C}$. Say

$$g(z) = c.$$

Then

$$f(z) = zg(z) = cz.$$

Conversely, if $f(z) = cz$, then

$$f(\lambda z) = c\lambda z = \lambda cz = \lambda f(z).$$

Therefore the desired functions are exactly

$$\boxed{f(z) = cz, \quad c \in \mathbb{C}.}$$

■

5. Let f and g be entire functions satisfying $f(0) = g(0) \neq 0$ and

$$|f(z)| \leq |g(z)| \quad \forall z \in \mathbb{C}.$$

Show that $f = g$.

Solution. Since $g(0) = f(0) \neq 0$, there exists $r > 0$ such that

$$g(z) \neq 0$$

for all $z \in B(0, r)$. Define

$$h(z) = \frac{f(z)}{g(z)}$$

on $B(0, r)$. Then h is holomorphic on $B(0, r)$. Also, by hypothesis,

$$|h(z)| = \frac{|f(z)|}{|g(z)|} \leq 1$$

for all $z \in B(0, r)$. Moreover,

$$h(0) = \frac{f(0)}{g(0)} = 1.$$

Thus

$$|h(0)| = 1.$$

So h attains its maximum modulus at the interior point 0. By the maximum modulus principle, h is constant on $B(0, r)$. Since $h(0) = 1$, we get

$$h \equiv 1$$

on $B(0, r)$. Therefore

$$f(z) = g(z)$$

for all $z \in B(0, r)$.

Since f and g are entire and agree on a nonempty open set, the identity theorem implies that

$$f = g$$

on all of $\mathbb{C}$.

■

1.2 Part II

6. Determine all entire functions that satisfy the functional equation

$$f(3z) = (1 - 3z)f(z).$$

Express the answer in terms of an infinite product.

7. Let $F(z)$ be holomorphic on a disc $D(z_0, r)$ so that

$$F(z_0) = F'(z_0) = 0 \quad \text{and} \quad F''(z_0) \neq 0.$$

Show that there are two curves Γ_1 and Γ_2 passing through z_0 that are orthogonal at z_0 and so that the function F when restricted to Γ_1 is real-valued and has a minimum at z_0 , and when restricted to Γ_2 it is also real-valued and has a maximum at z_0 .

8. Prove that if f is a non-constant bounded holomorphic function defined on the half plane

$$\{z : \operatorname{Im} z < 1\},$$

then there is some point on the real axis where the value of f is not real.

9. Show that there exists a sequence of polynomials (P_n) so that

$$\lim_{n \rightarrow \infty} P_n(z) = \begin{cases} 1, & \text{if } \operatorname{Re}(z) > 0, \\ 0, & \text{if } \operatorname{Re}(z) = 0, \\ -1, & \text{if } \operatorname{Re}(z) < 0. \end{cases}$$

Also, explain why (P_n) can't be bounded on compact subsets of $\mathbb{C}$.

10. Let $\mathfrak{F}$ be the collection of holomorphic functions on the open unit disk

$$D = \{z : |z| < 1\}$$

satisfying

$$\iint_D |f| \, dx \, dy \leq 1.$$

Prove or disprove that $\mathfrak{F}$ is a normal family.

August 2020

2.1 Part I

1. Let f be an entire function. Show that f is a constant function if there exist $M > 0$ and $R > 0$ such that

$$|f(z)| \leq M|z|^{1/2} \quad \text{whenever } |z| > R.$$

2. Fix $w = re^{i\theta} \neq 0$ and let γ be a piecewise smooth curve in $\mathbb{C} \setminus \{0\}$ from 1 to w . Show that there is an integer k such that

$$\int_{\gamma} \frac{1}{z} dz = \log r + i\theta + 2\pi ik.$$

3. (a) Let $a > 1$. Show that the equation

$$ze^{a-z} = 1$$

has precisely one root in the closed unit disk $\{z : |z| = 1\}$.

- (b) Show that the root in part (a) is positive.

4. Using the maximum modulus principle, prove the Fundamental Theorem of Algebra.

5. Let f be an entire function, taking real values on the real axis, and imaginary values on the imaginary axis. Show that $f(z)$ is an odd function; that is,

$$f(-z) = -f(z)$$

for all z .

2.2 Part II

6. Given entire functions f, g without common zeros, prove that there exist entire functions A, B such that

$$Af + Bg = 1.$$

Hint: Use the Mittag-Leffler Theorem.

7. If f is a nonconstant entire function, show that

$$(f^3 + f^2)(\mathbb{C}) = \mathbb{C}.$$

8. Suppose that G is a region in $\mathbb{C}$, $\{f_n\}$ is a sequence in $H(G)$, f is a nonconstant function, and $f_n \rightarrow f$ in $H(G)$. Let $a \in G$ and $b = f(a)$. Show that there is a sequence $\{a_n\}$ in G such that

(i) $a = \lim a_n$,

(ii) $f_n(a_n) = b$ for sufficiently large n .

9. Let

$$G = \{z : \operatorname{Re}(z) > 0\},$$

and $f : G \rightarrow G$ be a one-one analytic function from G into G . Suppose $f(a) = a$ for some positive real number a . Show that

$$|f'(a)| \leq 1.$$

10. Let $0 < R_1 < R_2$ and $\alpha \in \mathbb{R}$, and let G be the annulus centered at the origin with radii R_1 and R_2 . Let

$$z^\alpha = \exp(\alpha \log z),$$

where $\log z$ is a branch of the logarithm. Prove that z^α is analytic in G if and only if α is an integer.

Chapter 3

August 2024

3.1 Part I

1. Let G and Ω be two regions in $\mathbb{C}$, and $h : G \rightarrow \mathbb{C}$ be a one-one analytic function, and $f : G \rightarrow \mathbb{C}$ be a continuous function with $f(G) \subset \Omega$, and $g : \Omega \rightarrow \mathbb{C}$ be an

analytic function with $g'(w) \neq 0$ for any $w \in \Omega$. Show that f is analytic if

$$h(z) = g(f(z)), \quad \text{for all } z \in G.$$

2. Let D be the open unit disk. Show that no analytic function $f : D \rightarrow \mathbb{C}$ has the property that

$$|f(z_n)| \rightarrow \infty, \quad \text{whenever } |z_n| \rightarrow 1.$$

3. How many roots does the polynomial

$$z^9 + z^5 - 8z^3 + 2z + 1$$

have between the circles $\{|z| = 1\}$ and $\{|z| = 2\}$?

4. Prove Liouville's Theorem by considering the following integral

$$\int_{C_R} \frac{f(z)}{(z-a)(z-b)} dz,$$

for a bounded entire function $f(z)$, where $C_R = Re^{it}$ with $t \in [0, 2\pi]$, and $a \neq b$ with $|a| < R$ and $|b| < R$.

5. Let G be a region, and

$$G^* = \{z : \bar{z} \in G\}.$$

If $f : G \rightarrow \mathbb{C}$ is analytic, prove that $f^* : G^* \rightarrow \mathbb{C}$ defined by

$$f^*(z) = \overline{f(\bar{z})}$$

is also analytic.

3.2 Part II

6. Prove that a nonconstant harmonic function on a region is an open map.

7. Let G be a region and $f, f_1, f_2, \dots$ be elements of $H(G)$. Show that $f_n \rightarrow f$ in $H(G)$ if and only if for each closed rectifiable curve γ in G , $f_n \rightarrow f$ uniformly on γ .

8. Let G be a region and suppose $\mathcal{F}$ is normal in $H(G)$ and Ω is open in $\mathbb{C}$ such that $f(G) \subset \Omega$ for every $f \in \mathcal{F}$. Show that if g is analytic on Ω and is bounded on

bounded sets then

$$\{g \circ f : f \in \mathcal{F}\}$$

is normal.

9. Show there is an analytic function f on

$$D = \{z : |z| < 1\}$$

which is not analytic on any region G which properly contains D .

10. Find an entire function f such that for every integer $n \geq 1$, f has n as a zero of order n . Find the most elementary solution possible.

Chapter 4

January 2025

4.1 Part I

1. Prove the Fundamental Theorem of Algebra by completing the following two steps for a nonconstant polynomial

$$p(z) = a_0 + a_1z + \cdots + a_nz^n,$$

where $a_n \neq 0$ and $n \geq 1$.

(a) Show that there is a nonzero polynomial $q(z)$ such that

$$\frac{1}{zp(z)} = \frac{1}{zp(0)} + \frac{q(z)}{p(z)},$$

for $z \neq 0$ and $p(z) \neq 0$.

(b) By means of integration over the circle $C_R = Re^{it}$ with $t \in [0, 2\pi]$, show that $p(z_0) = 0$ for some point z_0 in $\mathbb{C}$.

2. Show that the equation

$$e^z = cz,$$

where c is a constant with $|c| > e$, has exactly one solution in the set

$$\{z \in \mathbb{C} : \operatorname{Re} z < 1\}.$$

3. Show that if $f : \mathbb{C} \rightarrow \mathbb{C}$ is a continuous function such that f is analytic off $[-1, 1]$, then f is an entire function.

4. Using Liouville's Theorem, prove the Fundamental Theorem of Algebra.

5. Let G be a region and $z_0 \in G$. Suppose $f : G \rightarrow \mathbb{C}$ is analytic and $f'(z_0) \neq 0$. Show that

$$\int_{\gamma} \frac{1}{f(z) - f(z_0)} dz = \frac{2\pi i}{f'(z_0)}$$

if $\gamma = \gamma(t) = z_0 + re^{it}$ for $t \in [0, 2\pi]$ and $r > 0$ is small enough.

4.2 Part II

6. Let G and Ω be domains in the complex plane. If $f : G \rightarrow \Omega$ is one-to-one and analytic and $\varphi : \Omega \rightarrow \mathbb{R}$ is subharmonic, show that $\varphi \circ f$ is subharmonic.

7. Let G be a domain in the complex plane and $u : G \rightarrow \mathbb{R}$ be a nonconstant harmonic function and let

$$A = \{z \in G : u_x(z) = u_y(z) = 0\}.$$

Show that A does not have any limit points in G .

8. Find an entire function f such that

$$f(ne^{2k\pi i/n}) = 0$$

for all positive integers n and all integers $0 \leq k \leq n-1$. Give the most elementary example possible.

9. Let G and Ω be two domains in the complex plane. Suppose $\{f_n\}$ is a sequence in $C(G, \Omega)$ which converges to f and $\{z_n\}$ is a sequence in G which converges to a point z in G . Show

$$\lim_{n \rightarrow \infty} f_n(z_n) = f(z).$$

10. Show that if $\mathcal{F} \subset H(G)$ is normal then

$$\mathcal{F}' = \{f' : f \in \mathcal{F}\}$$

is also normal.